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Improving picking performance at a large retailer warehouse by combining probabilistic simulation, optimization, and discrete-event simulation

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Abstract

Distribution warehouses are a critical part of supply chains, representing a nonnegligible share of the operating costs. This is especially true for unautomated, labor-intensive warehouses, partially due to time-consuming activities such as picking up items or traveling. Inventory categorization techniques, as well as zone storage assignment policies, may help in improving operations, but may also be short-sighted. This work presents a three-step methodology that uses probabilistic simulation, optimization, and event-based simulation (SOS) to analyze and experiment with layout and storage assignment policies to improve the picking performance. In the first stage, picking performance is estimated under different storage assignment policies and zone configurations using a probabilistic model. In the second stage, a mixed integer optimization model defines the overall warehouse layout by selecting the configuration and storage assignment policy for each zone. Finally, the optimized layout solution is tested under demand uncertainty in the third, final simulation phase, through a discrete-event simulation model. The SOS methodology was validated with three months of operational data from a large retailer's warehouse, successfully illustrating how it may be successfully used for improving the performance of a distribution warehouse.

Keywords: warehouse design; storage assignment policies; picking performance; discrete-event simulation; mixed integer programming; simulation-optimization

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1. Introduction

Warehouses are a critical component of modern supply chains, storing and buffering goods that arrive from suppliers located all over the world—sometimes continents apart—in single point, from which they are then sent to other locations, usually retail stores. Ideally, goods would be shipped just in time directly to the stores, or at least in such a way that stocks would be kept to a bare minimum, while keeping service levels at a satisfactory level. In practice, however, shipping times and transportation costs turn this option impracticable, or at least too expensive. Furthermore, they make it hard to deal with demand uncertainty, potentially leading to stockouts.

Arguably, the most labor-intensive and costly activity in a warehouse is order picking (de Koster et al., 2007; Rushton et al., 2014), which involves retrieving the requested items from storage to satisfy client orders. Drilling down the underlying tasks of order picking, Tompkins et al. (2010) estimate that roughly 50% of the order-picking time is spent traveling through the storage locations, 20% is spent searching for the item, 15% is spent for actually picking the item and moving it to the car or basket, and 15% for setup and other tasks. In addition, some authors estimate that order picking accounts for 60–70% of warehousing costs (Chen et al., 2015).

Although warehouse operations are technically simple to execute, improving warehouse efficiency entails many nontrivial decision-related issues. The warehouse manager faces many options regarding operational and tactical decisions when it comes to setting up a warehouse (de Koster et al., 2007). To name but a few, a diligent warehouse manager has to decide on the layout design for the warehouse and for each storage area (Sprock et al., 2017); define storage assignment policies, namely where items should be stored; consider picking policies, elaborating a list of which items and how many items to pick; implement routing policies, as the route taken may increase significantly the order retrieval time, especially in large and sparse warehouses; and so on (Davarzani and Norrman, 2015). Moreover, the warehouse manager needs to solve these problems while striving to minimize the travel distance/time and order throughput times (Grosse et al., 2017).

Addressing each of the said issues individually is already a difficult task, and even more so when we consider the integrated problem, as the impact of each of these policies is interdependent (van Gils et al., 2018b). For instance, no matter how optimized a picking policy is, if the layout design is flawed, it may compromise the picking performance, and *vice versa*. Moreover, and according to the literature, it is clear that there is not a single, one-size-fits-all solution exists; hence, optimal solutions can only be applied to very specific settings, and therefore, they are nongeneralizable. When all the degrees of freedom for arranging a warehouse are added up, the problem becomes increasingly complex. This is especially true given the onset of e-commerce, which has caused the stock-keeping unit (SKU) assortment to increase considerably (de Koster et al., 2017). Therefore, it is even more relevant to address warehouse problems in an integrated way (van Gils et al., 2018a).

This paper presents a three-stage methodology anchored on simulation and optimization that incorporates both zone layout design and storage assignment policies as warehouse management levers. Each phase solves a different part of the problem: the first finds the best storage assignment policy for each zone; the second considers the warehouse as a whole, and finds the best location for each zone with regard to throughput; and finally, stage three subjects the model to demand uncertainty. The model was tested on a picker-to-parts warehouse of a large distribution

company, achieving significant efficiency gains by decreasing the travel distance, as well as the order retrieval time.

The main contribution of this paper rests on the methodology hereby presented, and specifically on its ability to help designing a warehouse, or implementing storage assignment and picking policies. The methodology also encloses the following contributions: (a) we introduce a probabilistic simulation model to capture the travel time of different order sizes in a picker-to-parts warehouse with unidirectional aisles and (b) we introduce a novel mixed integer mathematical model to simultaneously define the configuration, location, and storage assignment policy in a multizone warehouse. Given the nature of the methodology, it can be easily extended to also study routing policies or other warehouse decision related problems, and can also be adapted to other multizone warehouses realities, by introducing the required changes to each phase without losing the methodological structure. The use of simulation facilitates subjecting the model to uncertainty, in this case to demand (order) uncertainty.

The remainder of the paper is organized as follows. Section 2 reviews the state of the art on warehouse science and order picking, while Section 3 presents the case study, along with relevant information. Section 4 explains the methodology employed, Section 5 presents the validation and the results obtained, and Section 7 concludes with a short discussion and further work.

2. Overview of warehouse decision problems

Warehouses are a sprocket of modern supply chains, helping to balance supply with uncertain demand (Gu et al., 2007). In an interconnected marketplace that serves customers from all over the world, surges in demand may happen frequently, with warehouses acting as a safety buffer (de Koster et al., 2007). Also, warehouses come as a compromise between transportation and operational costs, decreasing the first, but increasing the second, ideally in an uneven way. According to ELA/AT Kearney (2004), it is estimated that warehouses add about 20% to logistic costs, in return for the savings obtained in the transportation costs.

These fulfillment centers are essentially a “pass through” for the goods, an intermediary stage between procurement and distribution. The four main activities carried out at a warehouse are (a) receiving the goods, (b) putting away the goods either on buffers or storage locations, (c) order picking, and (d) shipping. The first two activities are inbound processes, while the two last are outbound processes (Bartholdi and Hackman, 2011). Order picking is arguably the most demanding and labor-intensive, especially in nonautomated, picker-to-part systems. In fact, most of the research on improving warehouse performance invariably rests on enhancing picker performance (de Koster et al., 2007).

In order to improve picking efficiency, warehouse managers typically face four tactical and operational decision related problems: (a) layout design, (b) picking policies, (c) storage assignment policies, and (d) routing policies (Chan and Chan, 2011). Layout design focuses on the disposition of the order-picking system, but also with the layout of the entire facility, including input and output (I/O) points, marshaling positions, proximity to cross-docks, etc. Picking policies define if and how orders are to be grouped into picking tours. Storage assignment policies specify how to store items through the warehouse. Finally, routing policies indicate the sequence of items to be picked.

Usually, items are sorted to ensure a single-pass retrieval, but exceptions exist. Each of the four decision problems is detailed next.

2.1. Layout design

Two decision problems occur when addressing the layout design (de Koster et al., 2007). First, how to design the facilities, including defining areas for loading and unloading docks, cross-docking, buffers, storage areas, marshaling positions (if any), etc. Second, how to define the layout within the order-picking designated area, also known as internal layout design or aisle configuration problem. Changes to the facility layout are outside the scope of this paper, and so we are mostly concerned with changes to storage areas. In this regard, Caron et al. (2000) presented an analytical approach for designing the picking area in low-level picker-to-part systems using a particular storage assignment metric, the cube per order index (COI), which takes into account the ratio between the product's total required space and the number of trips required to satisfy its demand per period. Petersen (2002) uses simulation to measure the effect of aisle length and the number of aisles on total travel time. Also, Larson et al. (1997) suggest a heuristic for assigning classes to locations, thus reducing the travel distance between I/O locations and aisle spots for the top-picked products. In a more recent work, Horta et al. (2016) propose a min–max mathematical programming model to assign stores to locations in order to minimize the distance traveled. The model is tested on a fruits and vegetables warehouse from a large retail chain.

2.2. Picking policies

In terms of options involving picking policies, the most common is a picker-to-part system, where the order picker walks or drives along the storage aisles to retrieve the items. Within this picking model, the decision over how to split and combine orders also matters, as it may severely affect the picking performance. For large orders with diverse products, each order can be handled individually, with each task giving origin to one picking tour. Alternatively, when orders are small, travel time may be reduced if different orders are grouped into one picking tour. The former is known as *single order-picking policy*, while the last is known as *batching* (de Koster et al., 2007). Clearly, single order-picking policy is administratively simpler, as no additional effort is required to split a batch and consolidate the items per customer order. Additionally, accumulation/sorting (A/S) processes are very prone to errors, especially when handled manually. Changes to picking policies are outside the scope of this work; therefore, we will skip a detailed review of this topic. The interested readers can refer to de Koster et al. (2007) for an extensive discussion on this subproblem.

2.3. Storage assignment policies

As regards storage assignment policies, alternatives abound, and they are known to influence significantly the distance traveled to complete an order. In general, it has been shown that nonrandomized, dedicated storage helps to maximize system throughput, while random storage is more

adequate when the goal is to maximize the utilization of storage space (Mansuri, 1997). Prior to actually defining where to assign products, space has to be allocated. One way of allocating the space within the storage area is the *forward-reserve* system. In this setup, the bulk stock (reserve area) is put on a place apart from the pick stock (forward area), thus reducing the picking area (de Koster et al., 2007). Holding everything else constant, the smaller the area, the lower the average travel times will be. However, dividing the inventory will require internal replenishments from the reserve to the forward area, and therefore a trade-off emerges. In van den Berg et al. (1998), the authors provide a model adapted from the knapsack problem to minimize the expected amount of labor during the picking period by optimizing replenishments. Öztürkoğlu (2020) proposed a biobjective mathematical model for allocating products in block staking warehouses. In this particular case, storage assignment targets two objectives: reducing travel distance and maximizing average storage usage.

With the picking and reserve locations well defined, the next decision concerns assigning products to particular aisles. The most common assignment policies are random storage, closest open location storage, dedicated storage, full turnover, and class-based storage (de Koster et al., 2007). Random storage consists in assigning a position that is selected randomly (i.e., with equal probability) from a set of eligible positions. This policy is known to promote high space utilization at the expense of increased travel distance, and is the method of choice for many of Amazon's distribution warehouses of low-volume SKUs (Weidinger, 2018). It also has the additional benefit of simplifying business rules. In a recent work, Ballestín et al. (2020) show how a multistage heuristic can be used for solving storage and retrieval problems in a random allocation setup, while taking into account congestion issues. For an evaluation of heuristics and exact methods in random assignment environments, refer to Petersen (1997).

In a similar fashion, the closest open location policy assigns items to the first available location. The nuance introduced in this method comparatively to the random assignment policy is concentrating the products next to the pick-up/drop-off areas, usually where the pickers start their tasks. This usually creates an uneven concentration of items next to I/O, which does not happen in the random assignment policy, except by pure chance.

On the contrary, nonrandomized policies, such as dedicated storage, assign products to fixed locations. This improves the picker's space awareness, as they become familiar with positions, while also improving their performance. However, space utilization is considerably lower, especially if there are many out-of-stock items. Full-turnover storage is a slight adaptation of dedicated storage, using the product turnover to assign items to the closest locations, while slow-moving products are stored in more distant locations. COI is frequently used as the proxy metric to turnover, and the rule consists of putting the lowest COI items closest to I/O locations (Heskett, 1963, 1964). If seasonality is removed from the equation, full-turnover storage would not differ much from dedicated storage. However, seasonal trends may cause products to change positions often during the year. The additional effort required to classify items is also not negligible, although modern information systems facilitate this process.

Finally, class-based storage divides products into classes according to a Pareto 20/80 distribution, ensuring that the fast-moving class contains about 20% of the products, but generating over 80% of the turnover. Methods such as ABC or COI-based class storage are common. The first method groups items into A-, B- and C-items, where A-items are the ones that generate about 80% of the turnover. B-items usually sit on the 80–95% range, and C-items are the remaining ones. The

particular nature of a warehouse warrants a discussion regarding the metric to use. Turnover may not fully convey the operational complexity added to picking, since very expensive, but seldom ordered items, may have high turnover, but do not generate many picking tasks. By occupying premium locations, they are taking out prime space that could be used instead for fast rotating items. Therefore, COI or pick volume is often used, as it translates turnover to demand frequency and picking effort more accurately. Either way, what follows is a discussion on how to position A-, B- and C-areas in low-level picker-to-part systems. Two possibilities are discussed in Jarvis and McDowell (1991). Regardless of the method chosen, the main idea is to bring these positions close to the depot points to minimize the tour's length.

Also worth pointing out is a subbranch of class-based storage methods that use the product family, or any other product properties to group together similar items. This area has evolved significantly in recent years, especially with the improvement in data mining methods. Perhaps the simplest method is to compare the statistical correlation between items, as proposed by Frazee and Sharp (1989) and applied, for instance, by Liu (1999), where it is shown that it can actually improve the efficiency of order picking. Several enhancements have then been proposed. Wutthisirisart et al. (2015) suggest an item-order relationship, using both order frequency and order size to generate an allocation algorithm to group together items. Zhang (2016) proposes several methods for this new branch of research called correlated storage assignment strategies. Two are sum-seed and static-seed clustering algorithms, and four other methods are sequencing algorithms for finding item sets. Following this strand of research, Battini et al. (2015) develop a multinomial probability model to define storage assignment policies at the microlevel, and warehouse design rules at the macrolevel.

2.4. Routing policies

The routing problem a warehouse manager faces inherits most of its properties from the traveling salesman problem, although certain singularities apply. First, congestion may be a problem (Bahrami et al., 2017), as too many pickers traveling through a corridor may hamper order retrieval time. But most importantly, there are thousands of picking locations, and as orders grow in size, so do the locations to visit. Consequently, exact methods grow exponentially in size, and are rendered useless as they cannot be solved in an acceptable time frame. For this reason, routing heuristics are used instead in practice (de Koster et al., 2007). Even though they may not guarantee optimality, they may generate solutions good enough to be used in practice in a reasonable time. By running multiple numerical experiments, Petersen (1997) estimates the best heuristic has a 5% gap from the optimal solution for a large number of instances.

Roodbergen (2001) provides an extensive review of the available heuristics. S-shape or traversal is one of the simplest and requires the picker operators to traverse the whole aisle in case there is one pick to be performed in that corridor. This leads to a serpentine-like path, with pickers moving back and forth along the aisles. In the return method, in contrast, the picker is allowed to make a U-turn and reverse his way. The midpoint method divides the storage into two, with front and back positions being accessed in two passes. Hall (1993) has shown that the midpoint method performs better when the number of picks per aisle is small (i.e., one pick on average), which depends not only on the size of the warehouse but also on the product assignment policy in use. The largest gap

policy does not differ much from the midpoint, except that pickers move as far on the aisle as the largest gap goes, meaning that they may cross the midpoint if required; moreover, and it has been shown that it always outperforms the midpoint policy (Hall, 1993). Finally, the combined heuristic uses a mix of S-shape routes and the largest gap.

In a more recent work, Lu et al. (2016) developed a dynamic routing algorithm with an extremely low running time that continuously reviews the best route and changes accordingly. The model is able to significantly reduce the order completion time.

3. Problem description

Our motivation comes from a distribution warehouse of one of the biggest Iberian nonfood retailers, aiming at improving overall operations efficiency, but also at optimizing the space being used, thus potentially avoiding, or at least delaying, an extra investment to move to a larger warehouse. This warehouse in particular features over 70,000 SKUs, mainly of sports and fashion goods, distributed through two different areas. In the first area, known as “Racks” there are five-level pallet racking systems, with ground-level locations being used for picking, and all the above locations assigned as reserve positions. The aisles may be visited by car only, and they are one way only, although shortcuts exist. The second area, labeled “Mezzanine,” is used for storing small items in cases of 50 cm × 50 cm. Picking is done manually, with operators walking around while carrying a box. Overall, including part- and full-time equivalents, there are around 120 professionals, working in eight-hour shifts, which is a small part of the 500 people in total who work at the distribution warehouse, with a significant number being assigned to administrative tasks.

Our work will focus on the “Racks” section, which currently features five different zones or business units (BUs). These zones contain articles ranging from sports footwear, sports clothing and equipment, nursery and clothing packs of multiple sizes, but also large and unusual formats such as treadmills. On an average day, the distance traveled by each picker on their cars is around 50 km. Considering that 120 pickers work at the warehouse, the total distance traveled over a period of 24 hours amounts to 3600 km.

Although the BUs share administrative resources as well as loading and unloading docks, the picking is actually performed by individual teams, and the orders never cross more than one zone. In fact, if it were not for the economies of scale gained from sharing administrative resources and loading docks, each BU could be located on separate warehouses. Order picking is fulfilled by a picking by store policy, meaning that picking orders are not aggregated to increase the quantity retrieved, except for some products that are picked on a batch and then delivered to an automatic sorter. Changing the picking policy was outside the scope of this project, and was therefore not considered, the same being true for the few SKUs picked as a batch and sent to the sorter.

Figure 1 illustrates the “Racks” section, as well as how the five different zones were initially distributed in the warehouse. Zones S1–S3 store sports department items, while zones F1 and F2 store fashion items. There are three fixed input/output (I/O) areas that work as drop-off points. Each zone has its own exclusive area, and they are not interchangeable. d_{SD} serves zones S1 and S2, d_{FD} is assigned to zone F1, and finally zones S3 and F2 use d_S as their drop-off point. Since thousands of picking tasks are performed each day, the distance traveled within the BU, or inner distance, may be as critical as the distance traveled back and forth to the drop-off points, or outer distance.

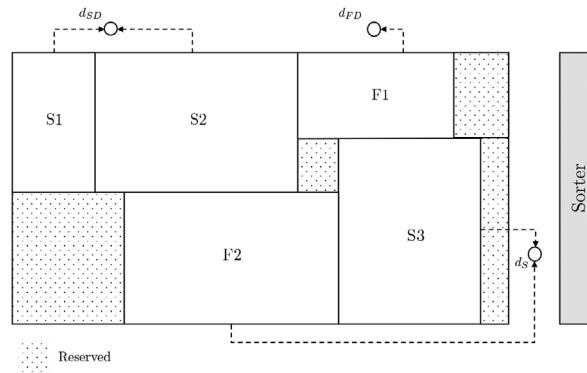


Fig. 1. Initial disposition of the five zones of the “Racks” section in the warehouse.

Finally, the “Reserved” section contains nonconformity items, such as treadmills or dumbbells, which are seldom ordered, and for which special picking guidelines exist. This section was excluded from the analysis.

The flow through each corridor is unidirectional and one way only, either north, that is, from downward to upward, or south, that is, from upward to downward, thus avoiding traffic jams caused by pickers traveling in different directions. Nonetheless, there are shortcuts that the pickers may take to cross the corridors without having to take the longest path. The BUs only define virtual areas for which a particular set of items is assigned.

The main challenge consisted in increasing productivity and reducing operational costs. Considering that about 50% of the order-picking time is spent on traveling between picking positions and drop-off points (Chan and Chan, 2011), and that no other changes (e.g., adding more drop-off points, building a new warehouse, changing the picking strategy, etc.) were to be considered, the approach would arguably require minimizing the distance traveled, both within the BUs, and from and to the drop-off points.

Therefore, the method to be devised for reducing the travel time spent on satisfying orders needs to address two separate, yet complementary problems: (a) within each BU, what is the best product assignment policy, that is, the one that minimizes the distance traveled to satisfy the orders placed; (b) considering that different BUs operate on the same warehouse, what is the best layout to accommodate the varying stress each BU generates? For the first question, we considered experimenting with different storage assignment policies, such as ABC-based, COI-based ABC, or random. With regard to the second question, the disposition and proximity to drop-off points of each BU had to be considered.

Our starting point is slightly different from the one usually reported in the literature, where 50% of the order-picking time is spent on traveling (Chan and Chan, 2011). In our case, we have found that 32% of the time is spent on traveling, and 68% on picking. If taken at face value, this could suggest that traveling time is already quite optimized or that picking is fairly inefficient, at least compared to the literature. In any case, only travel time will be considered as a performance metric to be improved. Additionally, we only report the work developed on the “Racks” section, without any loss of generality to the reader, since the methodology was also applied to the “Mezzanine” section quite similarly.

4. Methodology

This distribution warehouse serves several business areas with different product types and demands, subject to different seasonal cycles and with different teams operating independently. Notwithstanding, the warehouse operates as one single unit with some parts shared by all business areas, such as the docking stations and marshaling points, and cannot be managed separately. In fact, sometimes different business areas may serve the same retail shop, although operations at the distribution warehouse are managed separately until the pallets are loaded into a marshaling point, and then onto a truck. This adds an additional layer of complexity to the problem and requires a methodology that can decompose the problem at two different levels: (a) the BU area or zone level, considering its layout and storage assignment policy impact on the distance traveled within each zone; (b) the whole warehouse operation, and the impact of the warehouse layout design on the distance traveled between BUs and drop-off points.

Given these requirements, we devised a three-stage methodology called simulation–optimization–simulation (SOS) that explores both simulation and optimization to improve picking performance, namely by suggesting changes to the layout or to assignment policies. Considering the first (S), we use a probabilistic model to simulate different configurations for each BU. We consider the disposition (horizontal or vertical positioning), the picking flow, and storage assignment strategies (closest open position, random, ABC, COI-based ABC). Based on this, we estimate, for an increasing number of order lines, the time required to fulfill orders considering different setups (see column “Avg. inner distance traveled” in Table 3 for an example of the outputs of Stage 1). In the second phase (O), we feed the best results obtained in the first (S) to a mixed integer programming (MIP) model that builds the overall racks area layout by selecting a configuration, for each BU, in order to minimize the distance traveled at the warehouse level. This includes the distance that was not considered in the prior stage, such as moving from base to the BU area, and from the BU to the I/O points. Since some BUs are more solicited than others and put a different stress on the warehouse, the optimizer explores this trade-off to provide an optimal warehouse layout. As an output of this stage, we obtain different layouts ranked according to the overall distance traveled. Finally, the third simulation process (S) analyses both randomness and potential operational problems that had not been considered before, such as traffic jams in the corridors that are one way only. This model is a discrete-event simulation (DES) that replicates real operations of the warehouse, providing visual guidance while performing sensitivity analysis. Moreover, it simplifies the verification and validation of the methodology. We apply this model to the two best layouts obtained in the previous stage in order to see which one copes better with uncertainty.

Figure 2 illustrates how the SOS methodology is iteratively applied to construct new layouts and storage assignment policies. The first model uses the probabilistic simulation model to evaluate different flows, configurations, and storage assignment policies for each BU. Since each BU acts independently from all others, with their own particularities, a one-size-fits-all solution would be suboptimal. Expected travel distances and order retrieval times are estimated for a set of possible combinations of layout and storage assignment policies. Next, this information is passed onto the MIP model. The optimal configuration at the BU level can be suboptimal at the warehouse level, thus we need to consider the trade-off between sacrificing individual BU’s performance and maximizing the overall warehouse throughput. The optimizer balances the level of importance each BU puts on the warehouse, and runs accordingly. Finally, the warehouse layout needs to be tested

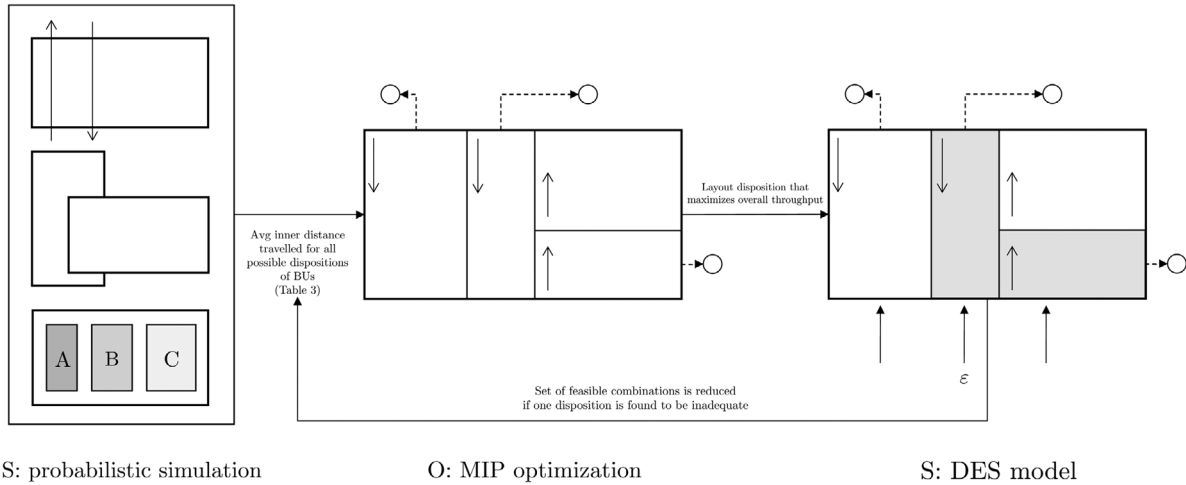


Fig. 2. How the SOS methodology can be applied to generate new layouts and storage assignment policies for a warehouse.

under uncertainty factors—that due to their complexity were not included in the previous steps. The DES model runs through several weeks’ worth of real data, as well as virtually generated data, to ensure the suggested warehouse layout and storage assignment policies perform as needed. If a particular BU disposition is found to be unfeasible or inadequate, we remove it from the list of available dispositions and run the optimization model again to redo the layouts. This feedback loop reduces the set of feasible layouts. We detail each of the stages next.

4.1. Notation

For the purpose of introducing our methodology, we start by describing the notation used throughout the paper. Consider Fig. 3 that depicts a 2D top view of one BU possible storage area. The layout is divided vertically into corridors and horizontally into levels. We index corridors using $c = \{1, \dots, C\}$ from left to right and levels using $k = \{1, \dots, K\}$ from top to bottom. Each corridor can only be crossed in one direction, either from top to bottom or otherwise. In the interception between a corridor and a level, we have an aisle $a \in \{1, \dots, A\}$ which are numbered from left to right and following the corridor direction. Considering the corridor’s direction, we can define that set D comprehending the aisles in corridors crossed from top to bottom, and set I composed of the aisles in corridors crossed in at the opposite direction. Each aisle’s length is l and the distance between the center of two adjacent corridors is e (as depicted in Fig. 3).

4.2. First stage: simulate the business unit

As mentioned before, the warehouse layout is split between several BUs or storage areas, each having its own independent team of pickers, with orders and teams of operators never crossing more

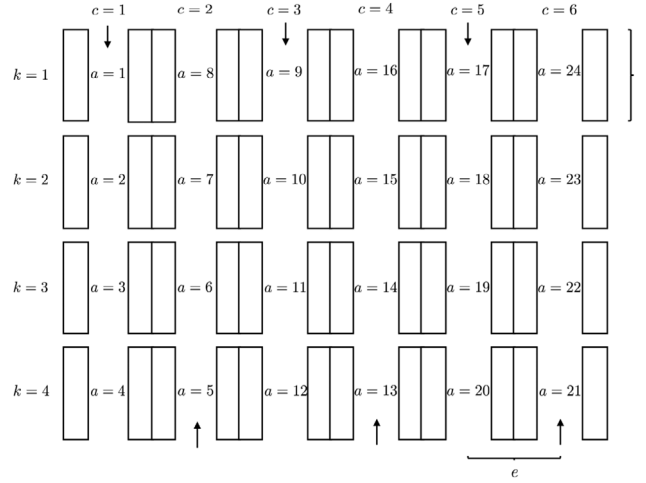


Fig. 3. A typical business area containing four levels and six corridors.

than one BU. Consequently, different storage assignment policies may coexist within the warehouse, and thus each BU can be analyzed individually.

Although picking tasks usually start with administrative and processing time, most of the time is spent on actually traveling between pick locations, and therefore improving picking efficiency requires minimizing the total picking time. Since we are leaving changes to the picking policy outside the scope of analysis, efficiency gains have to stem from reducing travel time between picking locations.

For this purpose, the BU's layout (i.e., number, size, and orientation of picking aisles) may be reconfigured, but also operating (i.e., sequencing items in the order list) and storage policies (i.e., item location in the storage area) may be changed. Changes to the physical disposition of aisles were outside the scope of the intervention, and the scope of analysis. However, the actual disposition of the storage area was considered, including the number of aisles and orientation (i.e., whether it expands horizontally or vertically).

In addition, storage policies were also questioned. We investigate the use of three alternative storage assignment policies: (a) closet open position, (b) random assignment, and (c) COI-based ABC categorization. The first in case distance correlated highly with order patterns, the last in case products were ordered randomly, with the benefit of avoiding congestion.

Considering these requirements, we developed a probabilistic simulation model, following the work of Caron et al. (2000). The model estimates the expected picking time inside each BU based on the order size (the number of SKUs), storage assignment policy, and BU configuration. The objective is to estimate the expected picking tour length $E[L]$ associated with a given task t , which can be written as

$$E[L] = \sum_{a=1}^A E[L_a], \quad (1)$$

where $E[L_a]$ is the expected contribution of aisle a to the total tour length. The total time can then be calculated using

$$E[time] = \frac{E[L]}{v} + n_t \cdot pick, \quad (2)$$

which adds up the time spent traveling between aisles, assuming pickers travel at an average speed of v , with the average time spent actually picking up the item ($pick$) multiplied by the total number of items n_t to be picked to complete the order-picking task t .

Let us suppose that there is an order to pick two items located on aisles a_2 and a_4 of Fig. 3. Using Equation (1) we compute the individual contribution of all aisles involved in the tour, which are a_1 through a_4 , plus a_5 – a_8 since the picker has to return to the I/O point. For the remaining ones not crossed, we flag them with a zero, thus providing a null contribution to the total tour length.

The expected contribution of any aisle depends on two factors: (a) its length and width; and (b) whether it has to be crossed or not. In the previous example, only aisles a_1 through a_8 have to be crossed; therefore, the chances of visiting other aisles are nil. The contribution of each aisle, $E[L_a]$, can be computed easily by adding up its length, l , and the distance between corridors, e , or in mathematical notation $(l + e)$. In the model implementation, these parameters are constant, as all aisles' and corridors' distance is the same. We can generalize this and define it as follows:

$$E[L_a] = (l + e) \cdot p(S_a). \quad (3)$$

The complicated part is computing $p(S_a)$, the probability of visiting aisle a , and therefore being part of the tour. There are only two reasons why aisle a needs to be crossed, and they are mutually exclusive: at least one item has to be picked from that aisle, which can be represented as E_a ; no items have to be picked from aisle a , despite being part of the route, which we can denote as V_a . In such case, $p(S_a)$ can be expressed as an arithmetic sum of the individual probabilities $p(E_a)$ and $p(V_a)$:

$$p(S_a) = p(E_a) + p(V_a). \quad (4)$$

Considering that a given task t has N items to be collected, we can define the probability of aisle a containing at least one item to be retrieved as follows:

$$p(E_a) = 1 - [1 - p(P_a)]^N, \quad (5)$$

where $p(P_a)$ is the probability that a pick addresses aisle a . With $p(E_a)$ well defined, we now need to calculate the odds of visiting aisle a even when no items have to be picked, $p(V_a)$. We can start by analyzing the case when the picker does *not* need to visit other aisles, that is, $p(V_a) = 0$, before reaching the next picking location. Such case only occurs when there is a direct move between the current picking position a and the next, a' . This is more common when the next aisle is positioned right after the current one (for instance, when an item has to be picked from aisle a_3 and then from

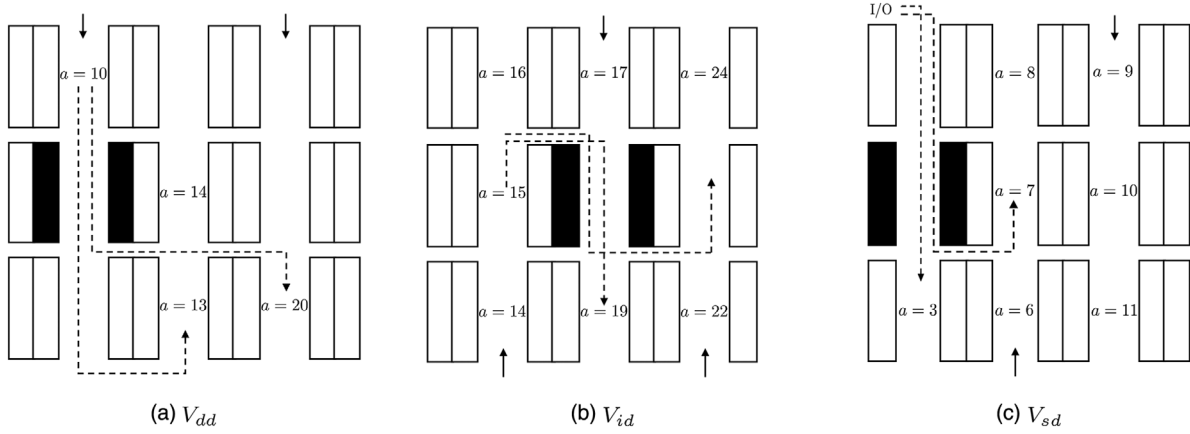


Fig. 4. Illustration of all the instances when a given aisle a needs to be visited on the way to the next picking location (for aisles belonging to domain D).

aisle a_4). We can provide a general expression for calculating the probability of going directly from aisle a to aisle a' without having to perform a picking task in between

$$p(E_a, E_{a'}) = \begin{cases} \left[1 - \sum_{j=a+1}^{a'-1} p(P_j) \right]^{N-1} \cdot p(E_{a'}), & \text{if } a < a' \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Shortcuts complicate matters further. Consider Fig. 3, where it is clear to see that aisle a_7 can be reached from a_2 by taking a shortcut. There are three cases in total that have to be addressed for calculating $p(V_a)$ when the visiting aisle is on an odd corridor (D):

$$p(V_a) = p(V_{dd}) + p(V_{id}) + p(V_{sd}), \quad \forall a \in D. \quad (7)$$

The first case we need to address (V_{dd}) is represented in Fig. 4a, and the purpose is to explore the cases when the next aisle in the corridor, for instance aisle a_{11} , may need to be visited, even though no picking takes place. If the picker is currently at aisle a_{10} and the next aisle is a_{20} , he can take a shortcut and go directly two levels up. The picker has not left domain D (aisles in odd levels). Now consider that the next item is on aisle a_{13} . Since the corridors are one way only, the picker cannot take a shortcut, and has to cross aisle a_{11} . In this case, it changes both domain (to I) and level. After generalizing, we obtain Equation (8). c' and k' will refer to the corridor and level of the next aisle to be visited:

$$p(V_{dd}) = \sum_{\substack{m \in D: m < a \\ k' < k \\ c = c'}} p(E_m) \cdot \sum_{\substack{j \in D \wedge k' > k \wedge c' \geq c \\ \vee \\ j \in I \wedge k' \geq k \wedge c' > c}} p(E_m, E_j). \quad (8)$$

The next case encompasses all instances when the next item to be picked is sitting on a corridor from domain D on the same level or above, or it is sitting on a corridor from domain I on the same

level or below (V_{id}). Figure 4b illustrates this case. Suppose the picker is currently at aisle a_{15} , and the picker's next assignment is at a_{19} . In such case, the next aisle in the corridor, a_{16} , does not need to be visited, since the operator can take a shortcut, but will have to pass through aisle a_{18} , as this aisle stands in the way. The same goes if the next item in line is on aisle a_{23} or any level below, in which case the picker also crosses aisle a_{18} . In any other case, the picker does not have to go through this aisle. The following Equation (9) generalizes these cases:

$$p(V_{id}) = \sum_{\substack{m \in I: m < a \\ k' \leq k \\ c' = c-1}} p(E_m) \cdot \sum_{\substack{j \in D \wedge k' > k \wedge c = c' \\ \vee \\ j \in I \wedge k' \geq k \wedge c' = c+1}} p(E_m, E_j). \quad (9)$$

Finally, we need to address the case when the picking task has just started, and the picker leaves from the I/O location (V_{sd}). Such case is illustrated in Fig. 4c. If the picker has to retrieve an item from aisle a_3 or from aisle a_7 , he must visit aisle a_2 as well. This occurs when the next item is on the same corridor, c , but one or more levels above ($k' > k$), or when it is on the same level or above ($k' \geq k$) but on the next corridor ($c' = c + 1$). Generalizing, we obtain Equation (10):

$$p(V_{sd}) = \sum_{\substack{j \in I \wedge k' \geq k \wedge c' = c+1 \\ \vee \\ j \in D \wedge k' > k \wedge c = c'}} p(E_0, E_j). \quad (10)$$

These three options cover all possible cases that imply crossing aisle a on the way to the next pick when the picker is currently on an aisle in the set D . We can now compute the probabilities of visiting aisles on domain I in a similar fashion. First, the probability of visiting aisle a on the way to the next picking position is generically given by

$$p(V_a) = p(V_{ii}) + p(V_{di}) + p(V_{ie}), \quad \forall a \in I. \quad (11)$$

Figure 5 illustrates the three cases when a given aisle a needs to be visited on the way to the next picking location, and the starting position is on an even corridor (I).

In the first case (V_{ii}), depicted in Fig. 5a, the picker has to visit aisle a_6 on the way to the next picking location, situated in the same corridor, but on a level below, or when it is in the next corridor, also on a level below. We cover these instances using the following expression, which is an adaptation of Equation (8):

$$p(V_{ii}) = \sum_{\substack{m \in I: m < a \\ k' > k \\ c = c'}} p(E_m) \cdot \sum_{\substack{j \in D \wedge k' \leq k \wedge c' > c \\ \vee \\ j \in I \wedge k' < k \wedge c' \geq c}} p(E_m, E_j). \quad (12)$$

The second instance (V_{di}), represented in Fig. 5b, covers the case when the current corridor is a direct corridor. As the illustration shows, if an operator is currently at aisle a_{15} and wants to reach an aisle on the next corridor, and on a level below ($k' \leq k$) or two corridors away on the same level

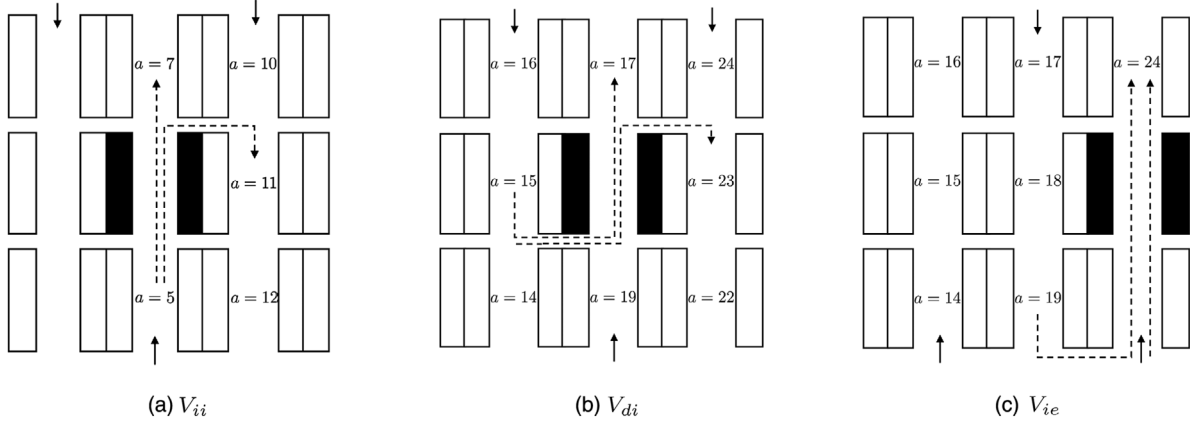


Fig. 5. Illustration of all the instances when a given aisle a needs to be visited on the way to the next picking location for aisles belonging to set I .

or lower level, then the operator has to cross aisle a_{18} on its way to the next aisle. We can generalize this using the following expression:

$$p(V_{di}) = \sum_{\substack{m \in D: m < a \\ k' \geq k \\ c' = c-1}} p(E_m) \sum_{\substack{j \in D \wedge k' \leq k \wedge c' = c+1 \\ \vee \\ j \in I \wedge k' < k \wedge c = c'}} p(E_m, E_j). \quad (13)$$

Finally, we adapt Equation (10) to cover the cases when an aisle a needs to be visited and the task is about to finish, which is represented in Fig. 5c (a_{24} is the last aisle). Suppose the operator is currently next to aisle a_{19} or next to the adjacent aisle, in this case aisle a_{22} . Hence, the operator has to go through aisle a_{23} to complete the task. In general, this happens when the last pick was on a D -aisle on the same or an higher level, or when it is in the same corridor and also on a higher level. The following expression translates these cases:

$$p(V_{ie}) = \sum_{\substack{j \in D \wedge k' \geq k \wedge c' = c-1 \\ \vee \\ j \in I \wedge k' > k \wedge c = c'}} p(E_j, E_0). \quad (14)$$

4.3. Second stage: optimizing the warehouse layout

As mentioned before, the warehouse comprises different BUs that operate separately, each having its own team of pickers and serving separate orders. For each BU, we have considered two levers for adjusting picking performance in terms of inner distance traveled: inner zone layouts (horizontal and vertical) and product assignment policies (ABC, COI-based ABC, random, etc.). The simulation model was helpful for obtaining key performance metrics for each BU, but each BU puts a different stress on the overall warehouse operations. Some BUs receive more orders than others, and the size of the teams also varies. Intuitively, it makes sense to privilege BUs more critical to

operations, when compared to those that put a lower burden on the operations, and bring them closer to I/O points, as movements back and forth drop-off points constitute a significant share of the travel time.

However, given the size of the warehouse, the possible combination of inner zone layouts and product assignment policies, and the number of BUs currently operating, the problem easily grows exponential in complexity. Back-of-the-envelope calculations are therefore pointless, with the problem requiring a more structured approach. In this sense, we developed an MIP model that tries to optimize the entire warehouse layout. It selects the best trade-off between minimizing the inner distance traveled inside each BU and the outer distance traveled to reach the I/O points. Obviously, this depends on the expected number of picking tasks each BU has, concerning the goal of minimizing the total travel distance of all the order pickers within the warehouse.

To introduce our MIP model, let $(x_z^{start}, y_z^{start})$ and (x_z^{end}, y_z^{end}) denote the input and output point coordinates for each BU indexed by $z = \{1, \dots, Z\}$ and let $T(n)_z$ be the expected total number of tasks having size n for BU z in the planning horizon. For each of these BU, the first-stage simulation model defines a set of possible configurations denoted by F_z . A configuration f can be described by the following parameters: h_f the height in aisles, l_f the length in aisles, $D(n)_f^{in}$ the expected inner distance of a task of size n , the expected entering $(x(n)_f^{start}, y(n)_f^{start})$, and exiting position $(x(n)_f^{end}, y(n)_f^{end})$ of a task of size n measured from the center of the upper left aisle of the layout. These configurations can also be grouped into two disjoint sets: (a) whose first aisle is a direct aisle (F^D) and (b) the ones with an indirect aisle as the first (F^I).

To define an optimal layout for the warehouse, we introduce the binary placement decision variables W_{fa} that are equal to 1 if the upper left aisle of configuration f is to be placed at aisle a , 0 otherwise. We also introduce the assignment decision variables Z_{fa} that are equal to 1 if the aisle a is assigned to configuration f , 0 otherwise. To correctly estimate the outer traveled distance, we first need to obtain the final entering and exiting position considering the arranged warehouse layout. To do so, let $X(n)_z^{start}$, $Y(n)_z^{start}$, $X(n)_z^{end}$, and $Y(n)_z^{end}$ be auxiliary variables that capture the expected entering and exiting position of a task of size n of BU z having into account the proposed layout. The outer distance traveled is then calculated by estimating the Euclidean distance between the entering and exiting point of the BU and its I/O point. Since the order pickers are unable to walk freely due to the aisle layout, this distance can be estimated by the sum of the differences in the x - and y -axis of the entering and exiting points to the I/O of the BU. Decision variables $\overline{Dx}(n)_z^{start}$, $\underline{Dx}(n)_z^{start}$, $\overline{Dy}(n)_z^{start}$, $\underline{Dy}(n)_z^{start}$, $\overline{Dx}(n)_z^{end}$, $\underline{Dx}(n)_z^{end}$, $\overline{Dy}(n)_z^{end}$, $\underline{Dy}(n)_z^{end}$ are introduced to capture the distance between the I/O and the entering (superscript *start*) and exiting (superscript *end*) points. The complete MIP model reads as follows:

$$\begin{aligned}
\min \sum_{z,n} T(n)_z \cdot (\overline{Dx}(n)_z^{start} + \underline{Dx}(n)_z^{start} + \overline{Dy}(n)_z^{start} + \underline{Dy}(n)_z^{start}) \\
+ \sum_{z,n} T(n)_z \cdot (\overline{Dx}(n)_z^{end} + \underline{Dx}(n)_z^{end} + \overline{Dy}(n)_z^{end} + \underline{Dy}(n)_z^{end}) \\
+ \sum_{z,n} T(n)_z \cdot \sum_{z,f \in F_z, a} D(n)_f^{in} \cdot W_{fa}
\end{aligned} \tag{15}$$

$$\text{s.t.} \quad \sum_{f \in F_z, a} W_{fa} = 1 \quad \forall z \quad (16)$$

$$\sum_{\substack{a': k' \in \{1, \dots, K\} \\ c' \in \{1, \dots, c-1\}}} Z_{fa'} \leq K \cdot C \cdot W_{fa} \quad \forall f, a \quad (17)$$

$$\sum_{\substack{a': k' \in \{1, \dots, k-1\} \\ c' \in \{c, \dots, C\}}} Z_{fa'} \leq K \cdot C \cdot W_{fa} \quad \forall f, a \quad (18)$$

$$\sum_{\substack{a': k' \in \{k+h_f, \dots, K\} \\ c' \in \{c, \dots, C\}}} Z_{fa'} \leq K \cdot C \cdot W_{fa} \quad \forall f, a \quad (19)$$

$$\sum_{\substack{a': k' \in \{k, \dots, K\} \\ c' \in \{c+l_f, \dots, C\}}} Z_{fa'} \leq K \cdot C \cdot W_{fa} \quad \forall f, a \quad (20)$$

$$\sum_{a'} Z_{fa'} \geq h_f \cdot l_f \cdot \sum_a W_{fa} \quad \forall f \quad (21)$$

$$\sum_f Z_{fa} \leq 1 \quad \forall a \quad (22)$$

$$X(n)_z^{start} = \sum_{f \in F_z, a} \left[(x_a + x(n)_f^{start}) \cdot W_{fa} \right] \quad \forall z, n \quad (23)$$

$$Y(n)_z^{start} = \sum_{f \in F_z, a} \left[(y_a + y(n)_f^{start}) \cdot W_{fa} \right] \quad \forall z, n \quad (24)$$

$$X(n)_z^{end} = \sum_{f \in F_z, a} \left[(x_a + x(n)_f^{end}) \cdot W_{fa} \right] \quad \forall z, n \quad (25)$$

$$Y(n)_z^{end} = \sum_{f \in F_z, a} \left[(y_a + y(n)_f^{end}) \cdot W_{fa} \right] \quad \forall z, n \quad (26)$$

$$\overline{Dx}(n)_z^{start} - \underline{Dx}(n)_z^{start} = x_z^{start} - X(n)_z^{start} \quad \forall z, n \quad (27)$$

$$\overline{Dy}(n)_z^{start} - \underline{Dy}(n)_z^{start} = y_z^{start} - Y(n)_z^{start} \quad \forall z, n \quad (28)$$

$$\overline{Dx}(n)_z^{end} - \underline{Dx}(n)_z^{end} = x_z^{end} - X(n)_z^{end} \quad \forall z, n \quad (29)$$

$$\overline{Dy}(n)_z^{end} - \underline{Dy}(n)_z^{end} = y_z^{end} - Y(n)_z^{end} \quad \forall z, n \quad (30)$$

$$X, Y, \overline{Dx}, \overline{Dy}, \underline{Dx}, \underline{Dy} \geq 0, \quad W, Z \in \{0, 1\}. \quad (31)$$

The objective function (15) aims at minimizing the total expected traveled distance at the warehouse level, presenting the sum of three terms: the outer distance traveled from the input point to the start of the BU area, the outer distance traveled from the end of the BU area to the output point and the inner distance traveled, respectively. Hence, the total expected traveled distance results from multiplying then total expected tour length (sum of outer and inner distances) of each order size, at each BU, by the number of expected tasks having such size. Constraints (16) ensure that exactly one configuration is selected for each BU from the potential candidates. The link between the BU configuration placement and the aisle assignment is guaranteed by constraints (17)–(20). The first two sets of constraints ensure that no aisle to the left (constraints (17)), nor above (constraints (18)) the upper left aisle selected, is assigned to the configuration. Similarly, the following two sets of constraints replicate the same logic to the right (constraints (20)) and bottom (constraints (19)), but having into account the size of the configuration. Constraints (21) ensure that the number of aisles assigned to the BU follow the selected configuration size and constraints (22), avoiding overlap between different BUs by not allowing the same aisle to be assigned to two different configurations. Constraints (23)–(26) calculate the coordinates of the expected entering and existing points for each BU depending on the task size. Finally, the constraint sets (27)–(30) estimate the outer traveled distance. Both constraints (27) and (28) estimate the outer distance in the x - and y -axis until the order picker enters the BU area, while constraints (29) and (30) estimate the distance traveled in these two axis after the order picker leaves the BU area. Note that two distinct variable sets are introduced to define positive (e.g., $\overline{Dx}(n)_z^{start}$) and negative (e.g., $\underline{Dx}(n)_z^{start}$) differences between the input/output point and the entering/exiting of the BU area.

4.4. Third stage: simulate the warehouse operation

The first stage dealt with the simulation at the BU level, generating estimates for the expected distance to be traveled when fulfilling an order for each possible configuration. In a second stage, the optimizer tries to combine all the possible configurations of inner layouts and product assignment policies for the different BUs, using the estimates obtained previously as inputs. This step is necessary since optimal solutions at the BU level may not generate optimal layouts at the warehouse level. Some BUs receive more orders than others, leading to different levels of stress to the warehouse operations. The optimizer takes into consideration this trade-off between sacrificing efficiency at the unit level but improving the overall result when generating a layout.

The third stage consists of a fully fledged discrete-event model that simulates actual picking orders in the layout suggested by the optimizer. The simulation model replicates the warehouse put-away and picking activities as close to reality as possible. There are pickers driving picking cars with speeds adjusted to match recorded speeds that fulfill orders in the exact same sequence as in real life. When completed or when the car is full, they drop the items at the assigned drop-off point, which may be a marshaling area or an automated belt. When the order is complete the operators return to I/O and are ready to fulfill another order. Otherwise, they go back to the BU and pick the missing items until the order is completed.

Three main activities had to be implemented in the DES model in order to reproduce the picking operations at the warehouse, along with other routine and setup operations, including drawing the layout or loading the SKUs from the database. The first activity concerns the logic behind

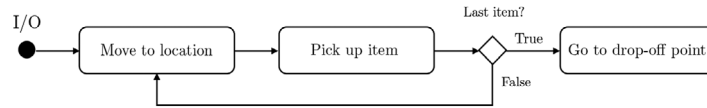


Fig. 6. State chart containing the order-picking sequence of steps.

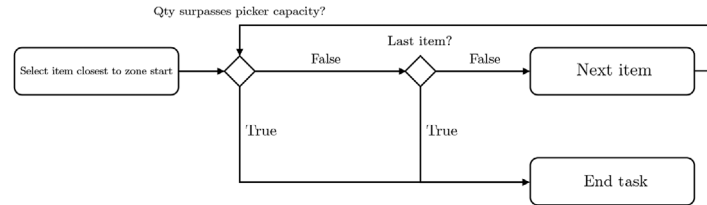


Fig. 7. State chart implementing the logic for creating picking routes.

order-picking tasks. The state chart depicted in Fig. 6 captures the steps necessary to successfully complete an order-picking task. The process is considerably simple since the complexity rests on sequencing the items to pick, which depends on the picking policy. Since the picking policy currently in place is not to be changed, the item list is retrieved directly from the information system's database. Hence, picking the items is a matter of moving the operator between picking locations until all items are retrieved from the order.

Ordering the items to be picked also requires devising a logical mechanism to guide the virtual operators, which we represent in Fig. 7. Since pickers follow S-shaped routes, the items can be ordered from the front to the rear of the BU according to the direction of the aisles, starting at the closest location to the I/O location. The task stops if either all items in the order are included in the item list or if picker capacity has been surpassed, in which case a new task will have to be created.

Finally, items arriving at the unloading docks need to be stored away. As it currently stands, items are assigned to the first available location. However, class-based storage has been introduced, and the state chart also takes into account that aspect. In case a picking location is already assigned, the newly arrived items go to reserve locations, which stand in positions above the ground level, in order to meet internal replenishment orders.

With all the logic in place, the DES starts by loading all the items and assigning them the exact same locations as in the real warehouse. Model parameters are defined at start up, and are listed in Table 1. Parameters regarding average picking time and average speed were manually derived from direct observation and measurement. Tasks are then initiated until all work for the day is completed, while simultaneously, items arriving through the unloading docks are put away. The model is then used to experiment with different inner layouts and storage assignment policies.

BUs are assigned in the warehouse, including orientation and direction of picking flows (corridors are one way only). The simulator then reproduces all the orders received within a time frame of several months and benchmarks it against the current layout. To do so, it assigns tasks to pickers following exactly the same rules used in the real warehouse, and tracks the distance traveled. In this stage, we can capture the impact of potential operational problems, such as traffic jams in the corridors. This is critical to guarantee a smooth operation, and may indeed require changes to the unit-level configuration, to the warehouse layout or to both, in a manual feedback process.

Table 1
Parameters and variables used in the discrete-event simulation model

Input	Characteristics	Frequency	Value
Storage assignment policy	Class of products that should go to each aisle	Startup	First free position, ABC
Products	Relevant characteristics of each product	Startup	Dimensions, weight, category, type, etc.
Drop point	Place where the picking tasks end	Startup	d_{SD}, d_{FD}, d_S
Average speed	Speed at which each picker travels	Startup	3 km/h
Picking time	Time to retrieve an item from a position	Startup	0.75–2.5 seconds
Number of pickers	Quantity of order pickers	Startup	120
Initial stock	SKUs and initial stored quantity	Startup	70k SKUs
SKUs received	SKUs that were stored on a given day	Daily	10–400
Orders	SKU and quantities ordered by each client	Daily	300–1000 orders, 11k–30k units

Finally, we study demand variability by simulating the warehouse operations for over 50 days of 24/7 work. The time period the data report to was chosen to be as variable as possible, including both periods of low activity (weekends), but also periods of extreme activity (one Black Friday). During these 50 days, the number of picks processed in the warehouse was 20.000, on average and on a daily basis, while the number of tasks was approximately 670 per day. However, there were peaks of 34.000 boxes and 1.091 tasks, almost twice the average processing throughput. In the same period, there were also slow days with 2.339 boxes collected and 100 tasks per days.

5. Results and discussion

In this section, we first validate the probabilistic simulation model and the DES based on a data sample composed of the real picking operations. Next, we show the application of the SOS methodology and identify the potential gains obtained using our suggested warehouse layout and storage assignment policies.

5.1. Model validation

Each step of the SOS method was validated independently, comparing simulated and actual data. Real data are very accurate, since picking operators wear a headset that connects to a voice picking system that informs the picker on the next position to visit, tracking every movement until the task is completed. The system is also used by management to obtain labor productivity metrics, including distance traveled and total time to complete a task.

For the probabilistic simulation model, we compared the expected time to complete a task by varying the number of picking operations. To do so, the model was run with several parameters, including number of groups of aisles and levels, aisle length, distance between groups of aisles,

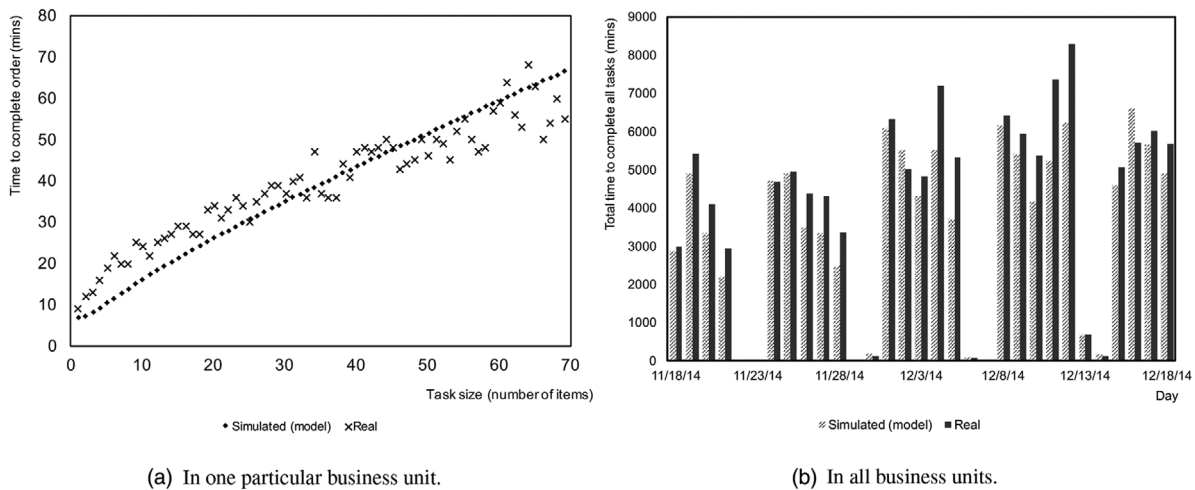


Fig. 8. Simulated versus actual data of the expected time to complete a task, depending on the number of picking operations.

average picker speed, and average picking time, all in accordance to the real warehouse setup. The size of the task (i.e., the number of items to pick) is then changed iteratively, and the expected time is compared with an average of the real order retrieval times.

Figure 8a presents the actual data against the times predicted by the probabilistic model. Using a standard measure to estimate the disparity between real and simulated values such as the mean absolute percentage error (MAPE), we can get an idea of how the model fares in estimating the time required to complete a picking order. In this particular case, MAPE is about 15.75%.

This average deviation may be explained by the fact that, in real life, picker cars may stop working, congestion occurs (we have not considered congestion in this model), pickers may let items drop to the floor, and so on. These delays could very well explain why the probabilistic model slightly undershoots the expected time to complete a task for orders up to 40–50 items. For larger orders, the model is more pessimistic. Visual inspection suggests that sometimes pickers cross aisles walking, particularly if the items to be picked are close to the end and to the beginning of the aisles, respectively. After a large number of picks such time difference accrues, reducing the picking time.

The DES model was validated using two different, but complementary strategies, typically used in simulation: verification and validation. First, we ensure that the model represents accurately the business processes in place in the warehouse, and that all relevant activities are executed as intended and match the specifications. For instance, we simulate one single agent fulfilling an order to verify if the proper sequence is followed; all the items are picked; and whether the items are dropped at the right drop point. This is performed for all the simulated operational activities of the warehouse.

Validation, on the other hand, determines the degree to which the simulation model and the data are an accurate representation of actual operations at the warehouse. In this sense, we validate the simulator by reproducing 50 straight days of real, nonrandomly generated tasks on the current warehouse layout, and compare those with the real statistics collected by the warehouse IT system. In Fig. 8b, we present one month of accumulated time for completing all the tasks in BU S2, which contains sports footwear, against the simulated time in the DES model. The MAPE is about 12.6%.

Table 2

Real and simulated times for 50 straight days of work in five different business units

Business unit	Real time (minutes)	Simulated time (minutes)	WMAPE (%)
F1	93,811	91,333	12.3
F2	123,886	126,247	14.0
S1	43,673	41,746	20.2
S2	199,148	183,343	12.6
S3	155,439	142,312	12.3
Total	615,957	584,982	13.2

The S2 area is the most demanding BU of the warehouse, as it deals with a considerable number of SKUs and picking orders. Nevertheless, other BUs also put stress on the warehouse operations. Table 2 provides an account of both real and simulated times for all BUs, as well as the deviation. The average weighted MAPE (WMAPE) is approximately 13.2%. The worst result obtained, 20.2%, represents a small BU that corresponds to approximately 7% of the warehouse picking time, and consists of picking very large items that are often difficult to place in the picking car.

5.2. Application of the SOS methodology

As a starting point, the probabilistic simulation model was to be used to generate the expected travel distance and order-picking time for all combinations of flows, layout configurations, and storage assignment policies (random, ABC, ABC COI-based, etc.). However, preliminary results showed that the COI-based ABC storage assignment policy Pareto-dominates all other policies, concerning all BUs. A detailed analysis of the data—after a data mining classification method—showed that a lot of items are frequently ordered together, and that seasonality is also a strong factor at play. These are strong points in favor of a nonrandom storage assignment policy, and more so toward a class-based policy, such as COI-based ABC.

Although this result cannot be generalized to all warehouses, it can be used to significantly reduce the number of combinations that need to be generated, thus reducing the solution space significantly, as only combinations of flow and layout disposition need now to be generated. Figure 9 shows the comparison between COI-based ABC and random assignment for one BU (same results apply to all others), the policy currently in place, with significant savings in terms of order retrieval time. For picking tasks containing between 10 and 30 items, up to 25% in time savings can be obtained using a COI-based ABC storage assignment policy. There is no task size for which a COI-based policy performs worse, so COI-based ABC is Pareto-dominant in the case under study.

Considering this, we have then generated all feasible combinations of flows and layout, which will lead to an optimal solution. For each BU, we have tried different feasible—not all BUs may be rotated, since the warehouse is rectangular—layout dispositions. We also experimented with different flows, including whether the operator enters the BU downward (N) or upward (S). This is required since the corridors are one way only, and therefore operators can only circulate downward and upward throughout the corridors, but it is possible to define the starting point of the path, forcing the person to start from the bottom or from the topmost position. For instance, a flow “S-N” means that the operator enters the BU from the position closest to the South, and therefore

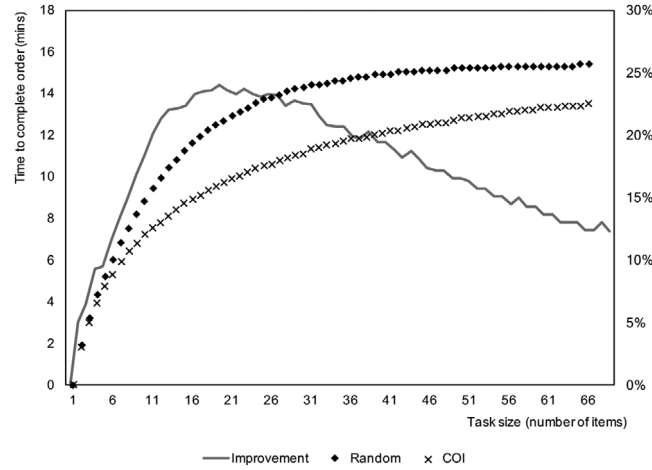


Fig. 9. Expected time to complete a task for one BU changing the storage assignment policy.

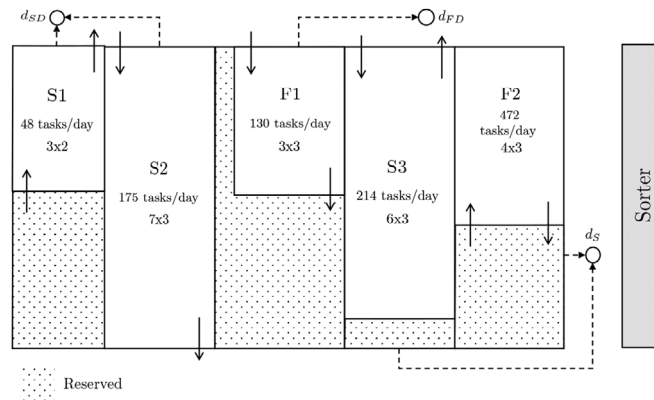


Fig. 10. Optimal layout that minimizes the total distance traveled, including inner distance (inside each BU) and outer distance (from and to I/O points).

moves upward; and that he or she leaves the BU from the uppermost position, moving upward. Finally, we have selected a task size for each BU that best represents the typical order-picking task by calculating a yearly average. Table 3 lists all the combinations that were simulated using the probabilistic model, and subsequently used as inputs in the MIP model.

Ideally, the best solution would be for each individual BU to be selected. However, such combination may not be feasible, as all BUs have to fit inside the current space available at the warehouse, causing a 2D packing problem. Moreover, the outer distance is also a strong factor at play (the most significant in this particular case, indeed), and therefore not all BUs may stay close to the I/O points. At that point, the MIP model that equates these trade-offs, and gives the optimal solution considering all constraints. It also takes into account the distance between each BU and the drop-off points, as well as the stress each BU puts on the warehouse. Figure 10 provides a graphical overview of the prescribed layout for our case study.

Table 3

Results obtained by running the probabilistic simulation model on a set of possible combinations

Business unit	Configuration (levels vs. corridors)	Flow (in, out)	Avg. task size	Avg. inner distance traveled (m)
S1*	3 × 2	S–N	28	349.5
S1 ⁺	3 × 2	N–S	28	280.2
S1	2 × 3	S–S	28	308.3
S1	2 × 3	N–N	28	464.9
S2 ⁺	11 × 2	N–S	80	783.1
S2	11 × 2	S–N	80	839.8
S2	7 × 3	S–N	80	911.8
S2*	7 × 3	N–S	80	798.9
S3 ⁺	8 × 2	S–S	70	542.6
S3	8 × 2	N–N	70	574.7
S3	16 × 1	S–S	70	506.0
S3	16 × 1	N–N	70	496.3
S3	6 × 3	S–S	70	616.4
S3*	6 × 3	N–N	70	717.5
F1 ⁺	9 × 1	N–S	26	285.6
F1	9 × 1	S–N	26	306.5
F1	5 × 2	N–S	26	346.8
F1	5 × 2	S–N	26	395.5
F1*	3 × 3	N–S	26	373.2
F1	3 × 3	S–N	26	474.8
F2	12 × 1	S–S	22	267.5
F2 ⁺	12 × 1	N–N	22	263.8
F2	6 × 2	S–S	22	321.5
F2	6 × 2	N–N	22	333.7
F2*	4 × 3	S–S	22	370.4
F2	4 × 3	N–N	22	420.9

Note: The current layout dispositions are marked with a +. The layout selected in the final optimized warehouse layout is marked with a *.

As expected, the optimizer proceeded by favoring BUs that put a higher stress on the warehouse (i.e., more tasks per day), moving them closer to the I/O points. In fact, all the inner layouts chosen by the optimizer perform worse in terms of average inner distance traveled when compared with the current configuration (cf. Table 3), but significantly better in terms of outer distance. Overall, the vertical layout improves overall performance (i.e., traveling time and picking time) by 5.6%, reducing traveling time by 18.8%. Although the impact in terms of traveling time is significant, its contribution to the total order-picking time is considerably smaller than the time spent picking up the items. Table 4 presents the results for each BU. With the exception of F1, for which no gains were obtained, all other BUs present significant gains in terms of traveling time.

However, more gains can be expected from also changing the storage assignment policy. Using again the DES model, we have tested the new layout configuration that the optimizer provided along with a COI-based ABC policy, and the improvement in terms of expected time of travel is slightly bigger. Table 4 shows the results obtained for each BU and for the whole warehouse operation in terms of time traveled and picking time. Figure 11 provides a heat map generated

Table 4
Impact of the change to the layout disposition of the warehouse

	Traveling time (minutes)				Picking time (minutes)				Total			
	Current	Alternative	Δ	Δ (%)	Current	Alternative	Δ	Δ (%)	Current	Alternative	Δ	Δ (%)
Horizontal layout with COI-based ABC storage assignment												
S1	112	104	8	7	663	663	0	0	775	767	8	1
S2	1284	970	314	24	2216	2217	–1	0	3501	3187	314	9
S3	913	815	98	11	1407	1406	1	0	2319	2221	98	4
F1	403	403	0	0	1452	1452	0	0	1803	1856	–53	–3
F2	764	529	235	31	1556	1555	1	0	2319	2084	235	10
Total	3476	2822	654	18.8	7293	7293	0.6	0	10,718	10,114	603	5.6
Vertical layout with COI-based ABC storage assignment												
S1	112	100	12	11	663	663	0	0	775	763	8	2
S2	1284	939	345	27	2216	2216	0	0	3501	3156	345	10
S3	913	582	330	36	1407	1407	0	0	2319	1990	330	14
F1	403	403	0	0	1452	1452	0	0	1856	1856	0	0
F2	764	515	249	33	1556	1555	1	0	2319	2070	249	11
Total	3476	2539	937	26.9	7294	7294	1	0	10,718	9835	883	8.2

Note: Although the vertical layout provides gains (approx. 2.6 percentage points over the horizontal (original) disposition), it requires a significant effort and operations downtime).

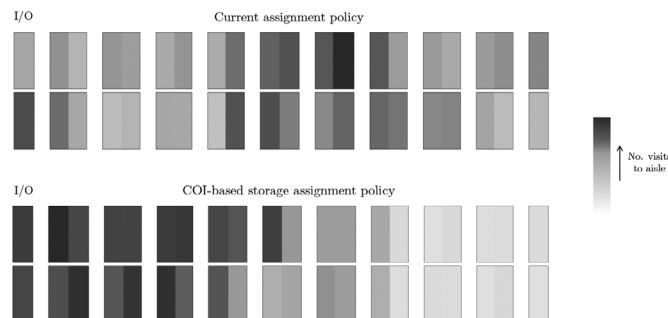


Fig. 11. A heatmap emphasizing aisles that were frequently and seldom visited (the darker the tone, the more visits). A COI-based ABC storage assignment policy concentrates picking on the region closest to the I/O point.

using the current storage assignment policy and simulated using a COI-based ABC policy for one of the BUs. As one can observe, a class-based storage policy concentrates frequently ordered items on the same space, which can be stored closer to the drop-off points. This reduces both the inner distance traveled, as C-items are less frequently picked, but also the outer distance, since pickers do not need to cross the whole BU and can leave anytime.

According to the results obtained, up to 26.9% can be gained in terms of traveling time from implementing a vertical layout along with a COI-based ABC storage assignment policy. Except for BU F1, for which there are no discernible improvements in the time traveled (similar to the previous case), all other BUs show strong gains in terms of picking performance. Its impact on the overall

operation is mitigated by the fact that traveling only accounts for 32% of the total order-picking time. As such, 68% of the time is spent on picking up the items and putting them in the car. Hence, the overall improvement is about 8.2%, which is still significant.

The SOS methodology allowed testing each of the changes, and involving the decision maker in the process by also providing a DES model where real-life operations can be tested. In fact, this has shown to be very important in earning the stakeholder's confidence. The case study illustrated its application in practice, and the results obtained show that it is a great tool to experiment new warehouse policies without having to disrupt operations.

6. Managerial insights

The SOS framework was the result of a consultancy-based research project that took place at one of the largest sports and fashion Iberian retailers. In addition to providing a real-case problem in which to test the model, the project was also a source of real data for validation. The distribution warehouse in which it was tested deals with over 70,000 SKUs, five different and independent BUs, and performs approximately 1000 picking tasks per day, amounting to thousands of picking lines.

Although the two scenarios that were presented to the steering committee originated savings, one required a sizeable downtime in the operations to be implemented. In the first scenario, the savings were estimated to be of around 5.6%, and required only the change of the assignment storage policy to a COI-based ABC. In the second scenario, which could lead to potential savings of 8.2%, it required a complete redefinition of the macro layout, which meant a downtime period of several weeks. These savings can also be converted into FTEs—which did not imply downsizing the workforce, in the particular case of this retailer, but rather a reallocation to other activities. The first scenario allowed for a reduction of 2.1 FTEs, while the second scenario could lead to savings equivalent to 3.1 FTEs.

Overall, the savings in terms of traveling time are much higher (18.8% for the first scenario and 26.9% for the second), but crossing picking locations amounts to no more than 32% of the total order-picking time in this warehouse, with the remaining time being spent on actually picking up the items. In fact, our results deviate considerably from the values reported in the literature, which typically report traveling as taking over 50% of the time Tompkins et al. (2010). If this were the case, the savings would be much higher.

As a result of this work, one of the first actions adopted by the warehouse managers was to implement the heat map, which provided them with a simple tool to measure the picking density over the warehouse and identify potential traffic jams. Most significantly, the firm went through with the COI-based ABC policy in the largest BU the one containing the footwear and the results gave confidence to the managers for scaling-up COI-based ABC to all BUs. Figure 12 illustrates actual results of the implementation of the COI-based ABC policy.

The second scenario, however, was put on hold until the next warehouse overhaul takes places, and this carries important lessons in terms of change management. Despite the promising results arising from the COI-based ABC storage assignment policy with the vertical layout, changing the actual layout of the warehouse meant bringing it to a full halt, in order to move the goods around and reorganize the pallet racking system. This downtime would incur in heavy costs to the organization, and a potential hit on the service level to the stores. When fully accounting for the benefits

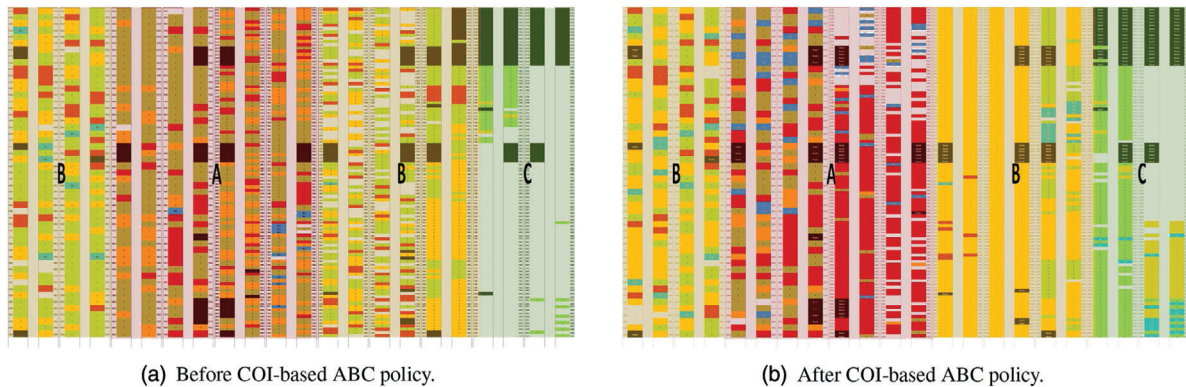


Fig. 12. Heat map of the footwear business unit before and after the implementation of the COI-based ABC storage assignment policy.

and the costs, the firm decided to wait for an opportunity to shut down the warehouse, and then proceed with the layout change.

7. Conclusion

Warehouse managers face many decisions, including the layout, policies related to storage assignment and picking, zoning, and also routing. When multiple BUs exist inside the same warehouse, decision makers face two levels of optimization, and hence an optimal solution at the lower, BU level may be suboptimal at the warehouse level. In such cases, an integrated approach to warehouse design and planning is advisable.

Considering this, our work proposes a novel SOS framework that provides insights at both micro- and the macrolevel, effectively assisting the decision maker in the planning of the warehouse. The three stages work as follows: first, a probabilistic simulation model estimates travel time inside each BU for different layouts; second, an optimization model uses the outputs of the probabilistic model for building the 2D packing layout, that is, combining the different BUs in order to minimize the travel distance of the overall operation; finally, a DES model is used to subject the new layout to a stochastic demand, observing whether the proposed configuration can cope with uncertainty. The SOS methodology can be used to guide the process of the warehouse design at the micro- and the macrolevel, even when no empirical data exists to estimate traveling time (one can use our first probabilistic simulation model to extrapolate times).

Although the policy levers considered only enacted changes to the configuration of each BU and to storage assignment policies, the SOS framework also makes it possible to test different picking and routing policies, requiring only slight changes to the DES model. For instance, the warehouse currently operates on a pick-by-store policy, but aggregated picking could also be put to the test using the simulator. This last stage is also useful for submitting the warehouse to different stressing conditions, including restocks and orders arriving at a random, stochastic order. This sensitivity analysis is crucial for finding an optimal layout that has an additional slack to cope with an array

of unanticipated situations. Finally, the DES model was very helpful for visually mapping solutions, thus helping decision makers understand the proposed changes and observe their impact.

As future work, changes to picking policies could also be pondered and incorporated in SOS. For instance, aggregate picking could be adopted instead of a pick-by-store policy. Such option would require, however, a place for sorting out the items through the different orders. The automatic sorter cannot be used for all items, so it would have to be a manual process, so far excluded. Two-level picking could also be considered. The four positions of the racks other than the ground level are used for reserve locations, but the first one could be assigned also for picking. Such option could reduce significantly the picking times, especially if frequently ordered times are grouped together. However, there are strong ergonomic restrictions to this option, since not all pickers have the height to reach higher levels, and some could actually injure themselves doing so. Either way, the SOS is extendable at any point, and can be used to study further enhancements to the warehouse.

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